SEBASTIAN ZAMBRANO

FULL-STACK & SYSTEMS SOFTWARE ENGINEER

Guayaquil, Ecuador | +593 99 813 4331 | sebastianzambrano2818@gmail.com | sebastianzambrano.dev | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Full-Stack & Systems Software Engineer with proven production track record designing, architecting, and delivering high-performance web applications, scalable SaaS systems, and low-level software. Core graphics engine developer for OmniNote (officially selected and indexed by the XERJ Open-Source Project in 2026 for autonomous AI code models). Specialized in React 19 / Next.js, strict TypeScript, PostgreSQL database design (RLS, indexing), and Linux systems programming. Focused on sub-16ms rendering latency, 95–100 Core Web Vitals, clean Git workflows, and robust API architectures.

TECHNICAL EXPERTISE

Core Languages: TypeScript, JavaScript (ESNext), C++20, Go, Python 3, SQL (PostgreSQL), Kotlin, GDScript, GLSL Shaders
Frontend & Web Architecture: Next.js 19 (App Router, SSR, SSG), React 19, Tailwind CSS, Vite, HTML5 Semantics, WCAG 2.2 AA Accessibility, Core Web Vitals Optimization
Backend, APIs & Databases: PostgreSQL, Supabase (RLS, Triggers), SQLite, RESTful APIs, JWT Authentication, Microservices, Asynchronous Processing, Redis
Systems & Low-Level: Linux Systems Programming, V4L2 Kernel Streaming (ioctl), C++20 Hardware-Accelerated Vector Graphics (Qt 6), TCP/USB Binary Protocols
Tooling & DevOps: Git, GitHub CI/CD, Docker, Vercel, Postman, Linux Terminal, Unit Testing, Agile/Scrum Methodologies

PROFESSIONAL EXPERIENCE

ESORIA Systems | Jan 2024 - Present

Full-Stack Software Engineer & Technical Lead

- Architected and deployed ContaNova, a multi-tenant commercial SaaS platform built with Next.js 19, TypeScript, Tailwind CSS, and PostgreSQL/Supabase, achieving 99.9% uptime in production.
- Engineered normalized relational database models, Row-Level Security (RLS) policies, and high-concurrency transactional queries handling real-time inventory and electronic invoicing.
- Designed and implemented automated REST services with digital XML signatures (XAdES), validation pipelines, and third-party API integrations.
- Delivered responsive web interfaces achieving consistent 98–100 Google Lighthouse scores across Performance, Accessibility, and SEO metrics.
- Led full software development lifecycle: architecture design, Git pull request reviews, automated CI/CD deployment pipelines on Vercel, and end-user documentation.

Actuariosa | Mar 2025 - Present

Systems & Data Automation Engineer

- Engineered an asynchronous Python high-throughput automation suite (MxCorreo) performing deep DNS resolution and MX verification across 132,000+ corporate records with zero bounce rates.
- Architected interactive financial analytics dashboards and data visualization tools using React and TypeScript for actuarial and risk modeling.
- Optimized data querying pipelines and batch database transformations, cutting data validation cycle times by over 70%.

XERJ Project & Open-Source Systems | 2025 - Present

Core C++ Systems & Engine Contributor

- Engineered a high-performance GPU-accelerated vector drawing core in C++20 and Qt 6, featuring sub-pixel Wacom digitizer sampling and real-time AST parsing.
- Vector stroke selection algorithms, spatial grouping routines, and comprehensive unit test suites were formally reviewed, accepted, and indexed by XERJ Project (2026) for AI coding models.
- Maintained strict memory safety, zero-copy buffer handoffs, and stable 60 FPS (sub-16ms latency) under intensive multi-thousand vector loads.

SELECTED SOFTWARE ENGINEERING PROJECTS

ContaNova SaaS | *Next.js 19, TypeScript, PostgreSQL, Supabase, Tailwind*

Commercial management and automated billing SaaS platform in production, featuring multi-business tenancy, real-time inventory tracking, and electronic validation.

OmniNote STEM Canvas | *C++20, Qt 6, SQLite, Hardware Acceleration*

GPU technical vector canvas with Wacom sub-pixel smoothing. Drawing engine core and test suite selected and indexed by the XERJ Open-Source Corpus Hub (2026).

DroidStream | *Go, Kotlin, Android CameraX, Linux Kernel V4L2*

Linux virtual webcam subsystem: standalone pure Go daemon with ioctl VIDIOC_S_FMT syscalls and custom binary streaming protocol (DSTR) over USB/Wi-Fi.

SwiftPiano | *Godot 4, GDScript, GLSL, MIDI*

Real-time 88-key polyphonic synthesizer featuring full QWERTY reactive mapping, custom GLSL additive shaders, and real-time live MIDI sheet transcription.

TechView Embedded AI | *Python, OpenCV, YOLOv8, Raspberry Pi*

Assistive embedded computer-vision edge system delivering real-time object classification and audio synthesis for visually impaired users. Awarded Institutional Graduation Honors.

EDUCATION

Undergraduate Degree: B.S. in Computer Science Engineering, UEES (In Progress - Nocturnal Mode)

Technical Degree: Technical High School Diploma in Informatics (Graduated with Honors)

ADDITIONAL INFORMATION

Work Eligibility: Remote Worldwide / Authorized to work in Ecuador

Languages: Spanish (Native); English (Intermediate / Professional Working B1-B2)

Code Repositories: github.com/FakeCxbas (100% auditable public code)

Live Portfolio: sebastianzambrano.dev (Live production web showcase)